Cricket Data Analytics

A comprehensive analysis of 3,099 ODI matches spanning 25 years, processing 1.64 million ball-by-ball records to uncover the game's greatest performers and strategic insights.

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Project Overview

3,099 Matches

Complete ODI internationals dataset from 2002-2025

1.64M Deliveries

Ball-by-ball records transformed from raw JSON files

2,505 Batters

Comprehensive career statistics computed

1,925 Bowlers

Full bowling analysis across all metrics

Built using Python, Pandas, and visualization libraries to create a complete analytics platform.

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Elite Batters: The Data Champions

Virat Kohli's Dominance

- 14,675 runs at 58.47 average
- 54 centuries and 75 fifties
- 91.96 strike rate
- 1,366 boundaries

Elite Quadrant

Only 5% of batters achieve average >45 AND strike rate >90. These elite players account for 30% of total runs.

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Top Run Scorers

14,675

Virat Kohli

India • 58.47 average • 54 centuries

11,640

KC Sangakkara

Sri Lanka • 46.19 average • 76 fifties

11,357

Rohit Sharma

India • 48.12 average • 32 centuries

10,274

MS Dhoni

India • 51.37 average • 72 fifties

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Bowling Excellence

1

Lasith Malinga

343 wickets • 27.75 average • 5.29 economy • 21 four-wicket hauls

2

Brett Lee

281 wickets • 22.59 average • 4.67 economy • Best strike rate: 29.05

3

Shakib Al Hasan

279 wickets • 29.57 average • 4.60 economy • All-rounder impact: 70.29

Death Over Specialist

Jasprit Bumrah: 5.21 economy in overs 45-50

Elite Bowling Quadrant

Economy <4.8 AND average <25: Muralidaran, McGrath, Pollock

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Elite Bowling Quadrant

Wickets

Malinga leads all-time

Strike Rate

Lee's best in class

Economy

Lee & Afridi's mark

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All-Rounders & Wicketkeepers

Shakib Al Hasan

Impact Score: 70.29

- 6,638 runs
- 279 wickets
- Only 12 players achieved 5,000+ runs and 200+ wickets

Wicketkeeping Elite

MS Dhoni: 426 dismissals • 118 stumpings (nearly double any other keeper)

KC Sangakkara: 399 dismissals • 11,640 runs

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Greatest ODI XI

Openers

Rohit Sharma • Quinton de Kock

Middle Order

Kohli • de Villiers • Sangakkara

Finisher

MS Dhoni (c/wk)

All-Rounders

Shakib • Watson

Pace Attack

Malinga • Starc • Brett Lee

Captain: MS Dhoni • **Team Balance:** Bats deep to #9 • **Bowling:** 3 pace, 1 spin, 1 pace all-rounders

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Phase Analysis Insights

Powerplay (1-10)

4.8 RPO • 1.2 wickets/match • Conservative start

Middle (11-40)

5.2 RPO • 1.8 wickets/match • Building phase

Death (41-50)

7.8 RPO • 2.4 wickets/match • Acceleration

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Key Findings & Future Vision

Elite Performance

5% of batters in elite quadrant (avg>45, SR>90) score 30% of total runs

Scout Identification

Quadrant analysis identifies undervalued players with elite potential

Match Strategy

Phase analysis informs powerplay and death over tactics

Future Enhancements

- Predictive modeling for match outcomes
- Interactive player comparison dashboard
- Real-time analytics and live predictions
- Web application for interactive exploration

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